Problem 2

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2.1 Long-run Elasticity

- In a long-run equilibrium, the price and consumption would not change over periods.
- $y_e = \frac{\alpha_0}{1-\alpha_1} + \frac{\beta_p}{1-\alpha_1} p_t + \frac{\beta_z}{1-\alpha_1} z_t + \mu_t$

2.2 Elasticity from Simple Static Model

- $y_t = \alpha_0 + \beta_p p_t + \beta_z z_t + \mu_t$

2.3 Elasticity Comparison

	Long-run		Static	
	Price	Income	Price	Income
Elasticity	-1.07	2.3	-0.84	2.12
Stand Error	0.56	0.35	0.025	0.048

2.4 Analysis

- It is very obvious to see that the absolute value of elasticity in the long-run is larger than that of static model.
- The economic intuition behind is that the stickiness of price and income in the long-run has vanished.
- People may need some time to react to the change in price or income. For instance, a person who always purchases one brand of milk may still purchase it as a custom without realizing the changing price. But he may buy a cheaper one later.